Reflection on Final Project

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Our final project remained largely consistent with the original proposal. The primary research question was to investigate the extent to which machine learning models exhibit measurable bias in hiring decisions and to evaluate how effectively fairness mitigation techniques can reduce bias without significantly impacting predictive performance, and it remained unchanged. The project continued to focus on fairness in AI-driven hiring systems, an area of growing importance as organizations increasingly rely on automated decision-making tools during recruitment.

One aspect that remained consistent with the proposal was the dataset selection. We used the *Hiring Discrimination in the Organisational Context* dataset which contains hiring-related survey responses and demographic information. This dataset enabled us to analyze fairness across multiple protected and vulnerable groups, including gender, ethnicity, parenthood status, relationship status, language proficiency, and educational background. The dataset also allowed us to explore intersectional bias by examining combinations of demographic characteristics rather than evaluating each characteristic independently. While the overall project goals remained unchanged, several aspects expanded beyond the original proposal during implementation. Initially we planned to focus primarily on gender and ethnicity. However, after exploring the dataset in greater detail we found additional demographic attributes that could be incorporated into the fairness analysis. As a result, we extended our evaluation to include parenthood status, relationship status, language proficiency, and education background. This provided a broader assessment of fairness and allowed us to identify disparities that may not have been visible when examining only gender or ethnicity.

The implementation process also revealed challenges that were not fully anticipated in the proposal. One significant challenge was class imbalance within the dataset as positive hiring outcomes represented a relatively small portion of the observations. To address this issue we incorporated class weighting and reweighing techniques to improve model performance and fairness evaluation. Interpreting fairness metrics required careful consideration because different fairness measures can lead to different conclusions regarding model behavior. Another area that expanded beyond the proposal was the depth of the evaluation. In addition to comparing Logistic Regression, Decision Tree, and Random Forest with SMOTE, and Gradient Boosting models we conducted a detailed fairness-performance trade-off analysis and compared multiple mitigation approaches across all stages of the machine learning pipeline. We implemented pre-processing (reweighing), in-processing (Exponentiated Gradient fairness constraints), and post-processing (threshold adjustment) techniques. During experimentation we expanded the project beyond the original proposal by incorporating SMOTE-based oversampling and Gradient Boosting. These additions improved model performance and provided a more comprehensive comparison between predictive accuracy and fairness outcomes. This allowed us to directly compare how different mitigation strategies affected both predictive performance and fairness outcomes.

Overall, the final project successfully achieved the objectives outlined in the proposal while providing a comprehensive fairness analysis than originally anticipated. The project demonstrated that machine learning models can exhibit measurable disparities across demographic groups and that fairness-aware mitigation techniques can reduce these disparities while maintaining comparable predictive performance. Through this work, we gained a deeper understanding of algorithmic fairness, the challenges of balancing fairness and accuracy, and the practical considerations involved in developing responsible AI systems for hiring applications.